An Empirical Evaluation of Hand-based Window Placement Techniques for Virtual Displays

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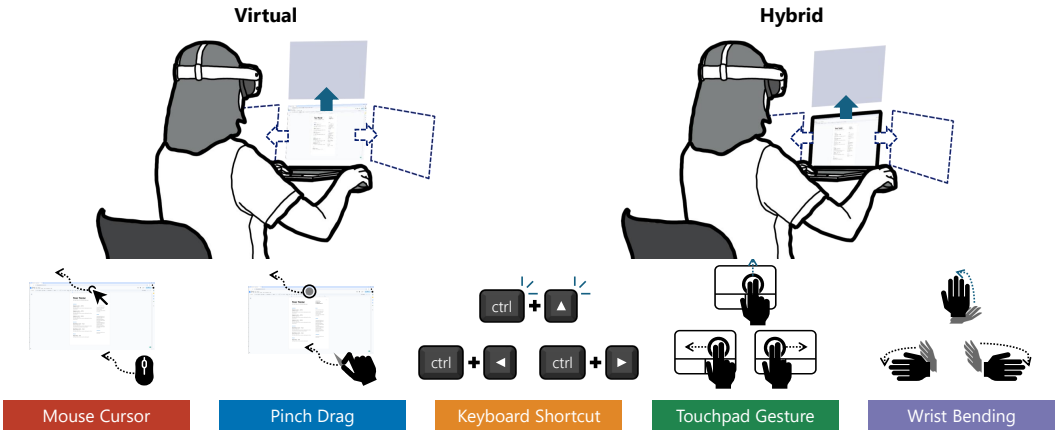


Fig. 1. This work evaluates five window placement techniques, including Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, and Wrist Bending, in both a fully virtual setup and a hybrid setup involving a laptop screen. This figure illustrates an experimental task in which the window is moved from the center to the top placement.

Existing research has demonstrated the benefits and envisioned broader applications of using extended reality (XR) head-mounted displays (HMDs) to create virtual display extensions for personal computers (PCs), offering a more flexible and cost-effective alternative to traditional physical monitors. This combined setup, with inputs from both sources available, provides users with more options for interaction. In this work, we aim to understand the advantages and disadvantages of these inputs and their interaction techniques in the fundamental window placement tasks in virtual displays enabled by the combined use of a PC and an HMD. With their standard inputs, we identified five potential hand-based window placement techniques: Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, and Wrist Bending. We conducted a mixed-design user study (N = 40) to evaluate user performance and experience with these window placement techniques in two display setups: a fully virtual setup where the HMD displays all windows, and a hybrid setup involving a

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laptop screen in addition to the HMD's virtual windows. All five techniques were evaluated in a structured layout task, while Mouse Cursor and Pinch Drag were additionally compared in a freeform placement task. Across both setups, Mouse Cursor was faster than Pinch Drag, while Keyboard Shortcut and Touchpad Gesture were faster and produced fewer failed trials than Wrist Bending. Laptop-based techniques also received higher usability ratings and lower perceived workload ratings than mid-air techniques. Nevertheless, some participants preferred mid-air techniques, valuing their naturalness. Our empirical findings can guide the future design of cross-device and cross-reality interactions between XR HMDs and PCs.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Mixed / augmented reality**.

Additional Key Words and Phrases: Extended reality, Head-mounted displays, Virtual displays, Interaction techniques, Window management, Multiple monitors, Empirical studies

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1 Introduction

Since the introduction of "windows, icons, menus, and pointers" (WIMP) graphical user interfaces, multiple windows on a screen have been enabled, revolutionized, and are now a fundamental feature of personal computers (PCs). When computer monitors became generally affordable, setting up a multi-monitor workspace became a common solution to provide more space for displaying additional windows or to reduce overlap among windows, thereby overcoming the limitations of a single monitor. With the increased bandwidth of visual information, a multi-monitor workspace offers several productivity benefits—it supports multitasking by allowing users to access more information simultaneously for their primary focus while maintaining peripheral awareness of secondary activities [13, 15]. This is particularly useful for knowledge workers, such as researchers, analysts, and programmers, who commonly need to handle large amounts of information. Today, these knowledge-based professions are no longer confined to their office; they now operate from various locations, including home, while traveling, or in the field [30]. With the shift toward more frequent changes in work locations and configurations, the use of traditional physical monitors becomes inconvenient. This issue underscores the need to develop a new solution to better support modern knowledge workers.

Extended reality (XR) head-mounted displays (HMDs) have advanced rapidly in recent years, becoming more powerful and lightweight. Through XR HMDs, users can interact with virtual content while also accessing the physical world. Such a blended environment makes an XR HMD a viable option for collaborating with a PC to create a multi-monitor workspace. Pavanatto et al. [28] defined a **virtual display** as an interface shown through an XR HMD that extends or replaces a PC's physical monitor, capable of displaying 2D windows and other content processed on a PC. Compared to physical monitors, virtual displays offer advantages in portability, cost-efficiency, flexibility, and privacy protection [29]. Ideally, a single HMD should allow users to have any number or types of virtual windows and be usable anywhere without taking up physical space. Previous research has hypothesized and validated the feasibility of virtual displays for productivity work [4, 17, 26, 29].

An essential need in a multi-monitor workspace is arranging windows. Recent work has developed novel techniques for window management in virtual displays. For example, Wang et al. [34] developed a hand gesture- and palm-based window management system in virtual reality (VR). Marguet et al. [23], on the other hand, used a mouse and keyboard combination as input to their XR window manager. We noticed that prior work has utilized different inputs for window management. The combined use of an XR HMD and a PC provides rich input options. On the HMD side, users

interact with spatial objects via controllers, bare-hand gestures, and eye or head gaze; on the PC side, a keyboard, mouse, and/or touchpad are typical input devices to support user interaction and productivity. There is a gap in **understanding how these standard and built-in inputs from an XR HMD and a PC support window management in their co-established virtual displays.**

This work takes the first step towards filling this gap by evaluating interaction techniques for **window placement**, one of the most essential elements in the window management workflow. We focus on hand-based inputs that are used most frequently in productivity work and evaluate five window placement techniques: Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, and Wrist Bending (see Figure 1). Among them, Mouse Cursor and Pinch Drag represent the fundamental drag-and-drop interaction strategy in PC and XR, respectively; whereas Keyboard Shortcut, Touchpad Gesture, and Wrist Bending use the input methods indicated by their names to provide shortcut-type interaction. We compared them in a purely virtual setup and a hybrid setup involving a laptop screen. We assessed their performance, usability, workload, and user preference when placing windows in a structured layout or freely. The results showed that Mouse Cursor enabled faster window placement than Pinch Drag in both the structured layout and freeform tasks across both display setups. In the structured layout task, Keyboard Shortcut and Touchpad Gesture were faster and resulted in fewer failed trials than the tested Wrist Bending implementation in both setups. Mouse Cursor, Keyboard Shortcut, and Touchpad Gesture also received higher usability ratings and lower perceived workload ratings than Pinch Drag and Wrist Bending. Nevertheless, participants' preferences suggest that these mid-air techniques may complement laptop-based techniques in window management.

This work mainly contributes to providing empirical findings of how essential hand-based inputs from a laptop and an HMD facilitate window placement in virtual and hybrid settings. These insights will be useful to designers and engineers in developing techniques and interactions that improve window management and productivity across devices and realities in the future.

2 Related Work

While a substantial body of literature has explored virtual displays, their interaction methods, and window management strategies, we specifically focus on those designed to support knowledge work scenarios in this section.

2.1 Working with Virtual Displays

Early in the 2000s, research has shown that having extra screen space can enhance task performance [9, 13, 15]. To overcome the physical space constraints, researchers and engineers have long considered virtual displays. With advances in XR technologies over the past decade, virtual displays powered by HMDs have become feasible and accessible. Pavanatto et al. [29] compared using an augmented reality (AR) HMD to display three virtual windows side-by-side for productivity work against three physical monitors and a hybrid combination of both with the same layout. They found that the physical monitors led to better performance and experience as expected, followed by the hybrid setup, and lastly, the fully virtual setup. The main drawback was the readability issues caused by the AR HMD's low FOV and low resolution. In their later work [26], they used a high-resolution VR HMD with lower texture distortion to display the virtual monitors and found that this reduced the gap between the physical monitors and showed benefits over a single laptop screen. Both studies examine the feasibility of using virtual displays, either extending physical monitors or replacing them entirely with virtual ones, and demonstrate an opportunity for use, especially when physical monitors are not achievable.

Unlike physical monitors, virtual displays offer greater freedom for users to place virtual windows. This leads to two threads of window arrangement strategies, both of which have been widely used

in the past literature: 1) users can freely place them within the virtual environment (e.g., [22, 23]); or 2) place them following a structured layout (e.g., [10, 24, 33]). Recently, Pavanatto et al. [28] compared the two approaches in purely virtual and hybrid setups and found that each has unique benefits and suggested their future use.

With the validation of virtual displays in laboratory environments, researchers have also explored their use in more challenging settings and tasks that are closer to real-world general use. Biener et al. [1, 3] have examined the long-term use of HMDs for knowledge work by replacing the physical desktop environment with a VR-based environment for one workweek. Although participants found the VR workplace uncomfortable and low in usability, they gradually became accustomed to the HMD and overcame negative feelings. Longitudinal studies using high-end HMDs, including Apple Vision Pro [6] and Varjo XR-3 [32], have received even more positive feedback from participants. Cheng et al. [8] conducted a two-week longitudinal diary study to examine factors that influenced users' usage patterns in an AR virtual display space, highlighting the use of AR workspace in the future, along with emergent hybrid workflows that involved physical screens and tasks. On the other hand, working with virtual displays in public environments is another center of attention. Pavanatto et al. [25] confirmed the acceptance of using XR HMD to accompany laptops in public settings, while Kaeder et al. [16] further explored the positions of virtual windows that make users feel safer and more accepted in public. Most recently, Büschel et al. [7] explored different window layout strategies for solving knowledge tasks in XR HMDs while moving between different public locations.

Given the above evidence, we envision that both the fully virtual and the hybrid monitor setups will be promoted and generalized in the future. This work considers window placement in both setups. In addition, the techniques are tested for moving windows following a structured layout and freely.

2.2 Interaction Methods in Virtual Displays

Common input devices for interacting with a PC, such as a mouse (and/or touchpad) and keyboard, can also be used with HMDs. The main challenge of using a mouse in HMDs' spatial environments is disambiguating its 2D inputs along the additional depth axis. Zhou et al. [36] solved this challenge by introducing *Depth-Adaptive Cursor* to infer cursor depth from scene context, enabling mouse-based pointing selection in 3D space and enriching mouse interaction with HMDs. A physical keyboard, on the other hand, is more single-purpose. It is mainly used for text entry tasks and is the most efficient method in HMDs [12, 18]. Since productivity tasks for knowledge workers heavily depend on the mouse and keyboard, this input combination has been utilized in virtual displays for knowledge work [4, 23, 35]. Mid-air hand-based interaction is the primary interaction paradigm for XR HMDs [19]. Current commercial VR HMDs use handheld controllers or track hand movements and gestures for interaction, while most AR or mixed reality (MR) HMDs only support hand tracking to ensure broader everyday use. Most work on HMDs uses hand-based interaction. For example, Lee et al. [20] investigated managing and interacting with windows in 3D space using a continuous hand gesture.

In this work, we concentrate on hand-based interaction techniques, which are the primary inputs for knowledge workers. There are also examples of using external devices to enhance handheld interaction, such as tablets [5], pens [2], and smartphones [31]. However, we choose not to expand the evaluation of these methods and instead focus on those available on current commercial devices. Therefore, we consider interactions supported by the mouse, keyboard, and touchpad on the PC side, as well as mid-air gestures on XR devices.

2.3 Window Management in Virtual Displays

Previous research has introduced window management systems for virtual displays utilizing various inputs. Biener et al. [5] presented a tablet-controlled window manager that allows the arrangement of multiple virtual windows around the user in immersive VR environments. Wang et al. [34] proposed *Handow*, a bare-hand gestural interface that enables direction manipulation of virtual windows on a user's palm. *WindowSpace*, introduced by Marguet et al. [23], uses traditional mouse and keyboard to handle window management in immersive 3D space.

There are several previous works that specifically focus on window switching. Shi et al. [33] introduce haptic retargeting to allow users to control and switch between multiple virtual windows in VR using an external touchpad. Pavanatto et al. [27] explored the *Spatial Bar* interface for window switching on virtual displays, where they found that using eye gaze via an HMD can facilitate window selection more than using a touchpad on a laptop, in particular with the cursor teleporting to the center of the window after selection.

To provide empirical evidence on input modalities for the future development of window managers, this work evaluates these modalities in window-placement tasks, which are fundamental to window management systems.

3 Methods

Our research question is: **how do different window placement techniques supported by current XR HMDs and PCs affect users' performance and experience when placing windows across fully virtual and hybrid display setups?** To answer this research question, we explore the strengths, limitations, and design implications of these techniques through an empirical user study to inform future cross-device and cross-reality interaction designs.

3.1 Testbed Environment

The testbed was informed by Pavanatto et al. [28] (as discussed in Section 2.1), albeit with modifications to focus on window placement.

We assume that a user has been focusing on their work in a maximized window on their primary display, which is directly in front of them. Now, the user wants to place this window to another location. The target location can be part of a structured layout or a random position. We define these two cases as:

- **Structured layout task**: the window is arranged according to a structured layout, which allows users to snap the window to the nearest system-registered location when moving close to it, or instantly relocate the window via command-based inputs.
- **Freeform position task**: the window layout is unstructured, and the snapping feature is disabled; users can place the window anywhere they want, but cannot perform automatic placement.

For experimental purposes, we simplify the testing task in two ways. First, although an actual system supports both placement types and executes them based on user input, we detach the two tasks in the testing environment and use them to evaluate the interaction techniques separately. Second, we only allow and ask participants to move the window from the center to its left, right, or top side, which are common positions and settings used daily. This controlled testbed helped us to isolate the testing variables and minimize confounding factors. The simplified tasks also reduce the learning cost for participants.

As mentioned, we used two display setups. The **Virtual** display setup has all windows exclusively in the HMD. The primary window is centered in the user's field of view and represents the one they focus on at the beginning of an experimental task. Participants were asked to move this

window to a target location—all happens in the virtual environment displayed by the HMD. In the **Hybrid** display setup, participants have a physical screen in front of them, surrounded by the virtual environment provided by the HMD. They need to move the window from the physical screen to the target location in the virtual environment.

3.2 Window Placement Techniques

As described in Section 2.2, there are various inputs existing in an HMD and a PC. In this work, we focus on the most fundamental ones, ensuring a wider applicability for current systems and learnability for most users. We compare five window placement techniques. **Mouse Cursor**, **Keyboard Shortcut**, and **Touchpad Gesture** are interaction methods using inputs from a PC. **Pinch Drag** and **Wrist Bending** are bare-hand interaction methods enabled by an HMD.

Mouse Cursor and **Pinch Drag** represent the most fundamental interaction languages for PC interfaces and XR spatial environments, respectively. They use the direct manipulation approach, which allows the user to act directly on the window via a visual indicator. Their interaction following the "Drag and Drop" sequence enables precise, freeform placement. They can be used to complete both structured layout and freeform position tasks.

- **Mouse Cursor**: To move a window, a user controls the mouse, which is shown as a cursor on the display, to locate the title bar of the window, presses and holds down the left mouse button, drags the window to the target place, and releases the mouse button to place the window.
- **Pinch Drag**: When a user wants to move a window, they first need to use their hand to either touch the title bar of the window, or ensure the ray emitted from the hand points at the title bar. They then perform a pinch gesture (thumb to index finger), move their hand while holding the pinch gesture for moving the window to the target location, and place the window by releasing the pinch gesture. Users can choose between direct touch and ray interaction depending on their preference and task needs.

The remaining three, **Keyboard Shortcut**, **Touchpad Gesture**, and **Wrist Bending**, by contrast, are command-based techniques for quick, automatic placement only. In other words, they can only be used to complete the structured layout task.

- **Keyboard Shortcut**: A user presses the keyboard shortcuts, usually a modifier key and an arrow key, to instantly snap the active window in the direction the arrow keys point. In this work, we set `Ctrl+Arrow` as the shortcuts.
- **Touchpad Gesture**: A user puts two fingers on the PC's touchpad and swipes on it to move the window to the corresponding locations.
- **Wrist Bending**: A user performs a mid-air gesture of bending their wrist to relocate the window to the corresponding direction of its current position. To prevent false triggering, we deliberately define an activation gesture. The wrist-bending gesture for moving the window only works when the palm of another hand is facing up.

The following section describes more how we implement these techniques in the testbed environment.

3.3 Apparatus and Implementation

A Meta Quest 3 VR HMD and a Lenovo Legion Y7000P laptop were the two main devices that enabled the interaction. Quest 3 has a 110° horizontal and 96° vertical field of view, a 2064×2208 per-eye resolution, a 90Hz refresh rate, and dual 18 PPD color passthrough cameras. The laptop features an Intel i5-12500H @ 2.50GHz CPU, an NVIDIA GeForce RTX 3050 Laptop GPU, and 32GB of RAM. It has a 15.6-inch screen that can open up to 180°. The experimental program was

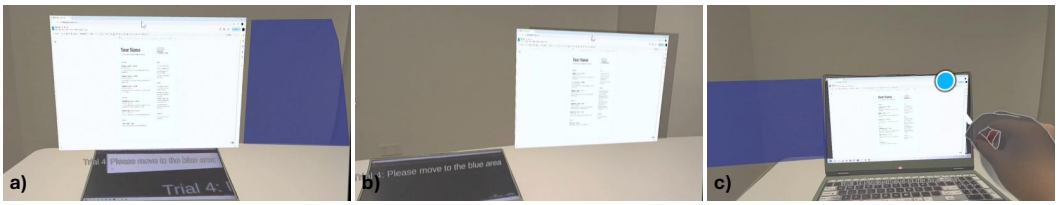


Fig. 2. a) The window placement task in the Virtual condition, asking the participant to use Mouse Cursor to move the window to the blue area. b) When the participant moves the window closer to the target location in the structured layout task, the target location turns gray to remind the participant that the window is going to snap to the target location. c) The beginning of a trial in the Hybrid condition. The participant is using Pinch Drag to move the window.

developed in Unity (version 2022.3.56f1c1) and was run on the laptop. The HMD was connected to the laptop via a USB-C 3.2 cable that supports a minimum of 5Gbps. All spatial elements were displayed in Quest 3's passthrough mode. We followed prior work [26, 29] to make this decision because video see-through VR HMDs typically have better display properties, especially a larger field of view, than current optical see-through HMDs. We opened the laptop screen to 180° to prevent overlap with the virtual monitors in the fully virtual display setup, and to 90° for use as the physical monitor in the hybrid display setup.

Mouse Cursor was empowered using a Logitech Pebble K380 wireless mouse, which was connected to the laptop via Bluetooth. The laptop's hardware components enabled Keyboard Shortcut and Touchpad Gesture, both of which were implemented via the operating system's native functionality. Pinch Drag and Wrist Bending, the two mid-air techniques, were tracked by Quest 3's built-in hand-tracking modules and were implemented using the Oculus Integration SDK (version 72.0). For Pinch Drag, we used the SDK's default index-finger pinch gesture without application-specific modifications. Wrist Bending was implemented as a custom technique. Activation occurs when palm of one hand faces upward, allowing an angular deviation of less than 30°. Window movement is triggered when the other hand's speed in the XY plane exceeds 0.5 m/s, with a minimum interval of 0.5 seconds between recognized gestures. Command detection is disabled when the activation pose is released or tracking data for either hand become invalid.

We used a screenshot of a typical text document interface to represent the window that the user was focusing on and aimed to move. In each experimental trial, the window was positioned in the center of the participants' field of view at a distance of 0.45m (Unity meters, the same hereafter). Its size was 0.4m W × 0.3m H in the Virtual setup and 0.333m W × 0.2m H in the Hybrid setup. The target position had the same size and depth, but was rendered in dark blue (#071E6ED6). It could be above (Virtual: 0.3m, Hybrid: 0.22m) or horizontally (Virtual: 0.42m, Hybrid: 0.35m) from the center window. Initially, we intended to keep all parameters consistent in the two display setups. However, we found the window positions looked strange and would be uncomfortable for users to conduct work in the Virtual setup (as they had to follow the laptop screen's configuration). Therefore, we set two groups of parameters to be in common, suitable workspace configurations, while ensuring they appear visually similar in scale. Due to the resolution of the simulated window, the title bar area was too small to interact with using Mouse Cursor and Pinch Drag. Thus, we set the height of the interaction area to 0.15 * the window's height, covering roughly the title bar and the ribbon area. In the structured layout task, the snapping feature was activated when the window was within 0.05m of the target location. The target position would turn gray (#00000080) as an indication. The snapping feature was disabled in the freeform task, so participants controlled the

movement throughout. The maximum offset for completing this task was 0.005m. Figure 2 shows the testbed environments.

3.4 Experimental Design, Task, and Procedure

We used a mixed design with two independent variables:

- A within-subjects variable **Technique** with five conditions: { Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, Wrist Bending }.
- A between-subjects variable **Display** with two conditions: { Virtual, Hybrid }.

Participants used five techniques to complete both structured layout and freeform position tasks in one of the display setups. They were asked to naturally move the window from the center to the target location, which could be on its left, right, or top side, without specific priority in speed or accuracy. Each direction had 5 repetitions, and they were given to participants in a random order. We placed text hints in the center of the scene to instruct participants to press the space key on the laptop to start the trial and to indicate when they had completed it. A 1-second interval was also set between two experimental trials. Participants sat at a table, wearing the HMD, and used the laptop placed on the table to complete the experiment.

Participants were assigned to a Display group before the experiment. At the beginning of the experiment, they first read and signed the consent form, then filled out a background questionnaire. Next, we introduced the participants to the experiment, including its objective, the equipment, interaction techniques, and tasks. Participants then began with structured layout tasks using five techniques. The order of the Technique conditions was counterbalanced using a balanced Latin-square design. Each Technique condition contained two sessions: a training session with at least 5 random trials, and a second session with 15 formal trials. The training session aims to help participants become familiar with the technique. During this session, participants could ask questions about the technique and practice more than the required trials until they felt they had mastered it. Performance data was only recorded from the formal trials. After each condition, participants completed questionnaires and took a break. Afterward, participants completed the freeform position tasks using the Mouse Cursor and the Pinch Drag techniques with the same procedure. The order of the two techniques was altered across participants. At the end of the experiment, we interviewed the participants to collect their feedback. The entire experiment lasted approximately 40 minutes.

3.5 Metrics

For each trial, we recorded the time and whether participants completed the task in one attempt. By "one attempt", we mean that the participants moved the window to the target location without any false triggers. False triggers occurred only with Keyboard Shortcut, Touchpad Gesture, and Wrist Bending. In our design, Mouse Cursor and Pinch Drag do not lead to failed trials because of their interaction mechanics. Therefore, we have completion time and number of failed trials as two objective measurements. For subjective measures, we employed a positive version of the System Usability Scale (SUS) [21] to gauge the perceived usability of the techniques in the given task. The ratings from 10 five-point items were converted into an overall SUS score (ranging from 0 to 100) for interpretation. We also used a NASA-RTLX questionnaire [14] to evaluate workload. An overall workload score was derived by averaging six subscale scores, each was scored from 0 to 100 in 5-unit increments. Additionally, we collected participants' feedback on their impressions of these techniques using a voice recorder. We asked them to comment on the techniques from two perspectives: (1) their strengths and weaknesses in completing two types of tasks, and (2) the

preferred combination when considering both tasks. The voice recordings of subjective feedback were transcribed into text for further interpretation.

3.6 Participants

We conducted an a-priori analysis using the G*Power tool [11] to determine the required sample size based on the planned mixed design with 2 groups (Display) and 5 measurements (Technique), as detailed in the Appendix A. The required total sample size was 38, yielding an actual power of 0.950. We decided to recruit 40 participants to ensure that the sample size was a multiple of both between- and within-subjects variables. This study obtained ethics approval from the institutional ethics committee.

Forty participants (22 females, 18 males) were recruited from a local university through social media and word of mouth. Their ages ranged from 18 to 37 years (M = 25.38, SD = 3.83). Four participants were undergraduate students, nineteen were master's students, fifteen were PhD students, and two reported that they were not students. Participants came from diverse academic backgrounds, including political science, financial technology, biometrics, computer science, robotics, etc. All participants had normal or corrected-to-normal vision. All except for one were right-handed. Twenty-nine participants reported using a computer for at least 6 hours on a typical weekday, eight for 4-6 hours, and three for 2-4 hours. Eight participants rated themselves as familiar or very familiar with XR HMDs, fourteen rated themselves as neutral, and eighteen rated themselves as unfamiliar or very unfamiliar with XR HMDs.

4 Results

During the experiments, we observed that several participants were distracted by the transition process from training to formal sessions and failed to perform appropriately in the first trial, so we excluded the first trial from each condition. To maintain consistent trial numbers across the target window's directions, we also removed the first trial from the remaining two directions in each condition. Thus, 2400 trials were used in the analysis of objective measures for the structured layout task (= 5 techniques × 2 display setups × 20 participants × 3 directions × 4 repetitions), and 960 trials for the freeform layout task (2 techniques instead). Given that each participant contributed multiple trials across conditions, we employed mixed-effects modeling to account for within-subject correlations and individual variability. The following subsections will detail the modeling for different measurements.

Throughout this results section, we use abbreviations for the five window placement techniques we tested: Mouse Cursor (MC), Pinch Drag (PD), Keyboard Shortcuts (KS), Touchpad Gesture (TG), and Wrist Bending (WB). We also abbreviate the structured layout task as SLT and the freeform position task as FPT whenever needed. The results were analyzed and visualized using R (version 4.5.2) and RStudio (version 2026.06.0).

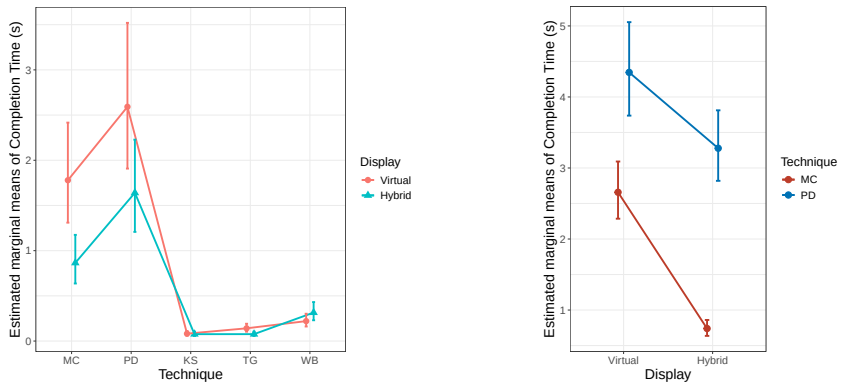
4.1 Completion Time

4.1.1 Structured Layout Task. Before analyzing completion time, we examined potential confounding effects, including: (1) the Trial Order that may lead to learning effects across trials, (2) the Direction of the target position, which was not treated as an independent variable in the experimental design; and (3) the carryover effect of Condition Order. A preliminary analysis using linear mixed-effects models found that Condition Order did not significantly affect completion time but revealed significant effects of Trial Order, Trial Order × Technique, Direction, and Direction × Technique. Therefore, we decided not to aggregate completion times across trials. The linear mixed-effects model was conducted on log-transformed completion times at the trial

level, with Technique and Display modeled as fixed effects, participants as a random intercept, and Trial Order, Direction, and their respective interactions with Technique as covariates.

As in the preliminary analysis above, Trial Order significantly affected completion time, and interacted with Technique, indicating different learning patterns across techniques. Direction also affected performance with a significant Direction × Technique interaction, suggesting direction-dependent performance differences. Detailed analyses of these effects are provided in Appendix B. After accounting for trial progression and window movement direction, the model revealed a significant main effect of Technique, $F(4, 2275.58) = 633.96$, $p < .001$, indicating differences in interaction speed across window placement techniques. The Technique × Display interaction was also significant, $F(4, 2275.57) = 13.24$, $p < .001$, suggesting the relative speed performance of techniques differed between display conditions. However, the main effect of Display was not significant $F(1, 40.04) = 2.61$, $p = .114$.

Estimated marginal means were calculated from the linear mixed-effects model and back-transformed to seconds for interpretation. Post-hoc pairwise comparisons with Tukey adjustment across techniques yielded similar trends under Virtual and Hybrid display conditions. The three command-based techniques, KS, TG, and WB, resulted in significantly shorter completion times than MC and PD (all $p < .001$). MC was significantly faster than PD (Virtual: $p = .011$; Hybrid: $p < .001$). KS and TG were significantly faster than WB ($p <= .002$). KS was faster than TG in the Virtual condition ($p < .001$) but not in the Hybrid condition. On the other hand, the Hybrid condition led to significantly shorter completion times than the Virtual condition for MC ($p = .001$), PD ($p = .039$), and TG ($p = .006$), while no significant display differences were observed for other techniques.[1] Figure 3a summarizes these results. Descriptive statistics are summarized in Table 1.



(a) Technique × Display interaction effect in SLT (b) Technique × Display interaction effect in FPT

Fig. 3. Estimated marginal means of completion time in seconds, with error bars indicating 95% confidence intervals. Both (a) structured layout task and (b) freeform position task show a significant Technique × Display interaction.

4.1.2 Freeform Position Task. As in the structured layout task, we examined confounding effects before analyzing completion time in the freeform position task and only observed a significant main effect of Trial Order. Given this, we included Trial Order as a covariate in the final model so that the effects of main variables were estimated while accounting for trial-related variability.

[1]Holm adjustment was used here and whenever only two conditions were compared within each subset.

Table 1. Descriptive statistics of completion time (seconds).

Task	Technique	Display	*M*	*SD*
SLT	MC	Virtual	2.57	1.61
	MC	Hybrid	1.95	1.46
	PD	Virtual	3.83	2.28
	PD	Hybrid	2.58	2.13
	KS	Virtual	0.31	0.28
	KS	Hybrid	0.30	0.40
	TG	Virtual	0.50	0.44
	TG	Hybrid	0.21	0.14
	WB	Virtual	0.66	1.02
	WB	Hybrid	0.69	0.52
FPT	MC	Virtual	3.27	1.35
	MC	Hybrid	1.17	0.72
	PD	Virtual	5.67	2.60
	PD	Hybrid	4.02	1.50

The final linear mixed-effects model was also conducted on log-transformed completion times at the trial level.

The analysis revealed significant main effects of Technique, $F(1, 920) = 521.65, p < .001$, and Display, $F(1, 40) = 67.55, p < .001$. Furthermore, the Technique × Display interaction was also significant, $F(1, 920) = 132.76, p < .001$. Estimated marginal means were back-transformed to seconds for interpretation. Post-hoc pairwise comparisons using Holm adjustment showed that MC was faster than PD in both display setups ($p < .001$), as shown in Figure 3b. Meanwhile, both MC and PD needed longer time to place windows in the Virtual condition than in the Hybrid condition (MC: $p < .001$; PD: $p = .010$).

4.2 Failed Trials in Structured Layout Task

As noted in Section 3.5, MC and PD cannot produce failed trials. Accordingly, only KS, TG, and WB in structured layout task were included in this analysis. Because failed trials represent binary outcomes, we adopted generalized linear mixed-effects models. Given the sparse and uneven distribution of failed trials across conditions, we examined potential confounding effects of Trial Order, Direction, and Condition Order through descriptive inspection before fitting the final model. The inspection revealed no obvious patterns related to trial progression, window movement direction, or the order of technique conditions, so these confounding factors were not included in the final model.

We fitted a binomial generalized linear mixed model to examine the effects of Technique and Display on the probability of a failed trial, with participants as a random intercept. The model revealed that Technique significantly influenced the probability of failed trials, $\chi^2(2) = 61.48, p < .001$, while the effect of Display was not significant, $\chi^2(2) = 0.97, p = .325$. Pairwise comparisons with Tukey adjustment showed that WB was associated with significantly higher odds of failed trials compared with KS ($OR = 14.66, p < .001$) and TG ($OR = 105.26, p < .001$). The difference between KS and TG was not significant ($p = .157$).

Note that the Technique × Display interaction was not examined in the model because their combinations contained no failed trials, resulting in unreliable estimation of interaction effects. As

Table 2. Observed failed-trial counts and rates across techniques (reported as failed trials / total trials (%)) in Virtual and Hybrid display conditions and estimated probability of failed trial with 95% confidence intervals (CI).

Technique	Observed counts (rates)		Estimated probability (95% CI)
	Virtual	Hybrid	
KS	6/240 (2.5%)	1/240 (0.42%)	1.01% (0.44%–2.31%)
TG	0/240 (0)	1/240 (0.42%)	0.14% (0.02%–1.04%)
WB	41/240 (17.08%)	36/240 (15%)	13.05% (9.11%–18.35%)

Table 3. Descriptive statistics of SUS scores (the higher, the better) and NASA scores (the lower, the better).

Task	Technique	Display	SUS Score		NASA Score	
			M	*SD*	*M*	*SD*
SLT	MC	Virtual	86.63	11.39	8.79	6.2
	MC	Hybrid	83.75	16.55	9.83	9.29
	PD	Virtual	69.63	14.92	18.13	12.45
	PD	Hybrid	72.88	18.63	18.13	14.54
	KS	Virtual	90.13	10.15	4.71	4.13
	KS	Hybrid	86.5	12.81	8.75	10.78
	TG	Virtual	85.25	13.74	6.29	4.56
	TG	Hybrid	93.75	9.55	4.38	5.02
	WB	Virtual	56.5	24.54	25.25	16.51
	WB	Hybrid	67.63	25.94	20.83	18.62
FPT	MC	Virtual	87.13	12.81	10.54	9.21
	MC	Hybrid	87	15.89	10.67	9.49
	PD	Virtual	64.13	17.7	24.58	13.52
	PD	Hybrid	75.5	20.66	20.5	17.39

a result, we inspect the interaction descriptively and summarize the results in Table 2. The higher failure rate of WB was observed in both Virtual and Hybrid conditions, suggesting that WB was error-prone regardless of display setups.

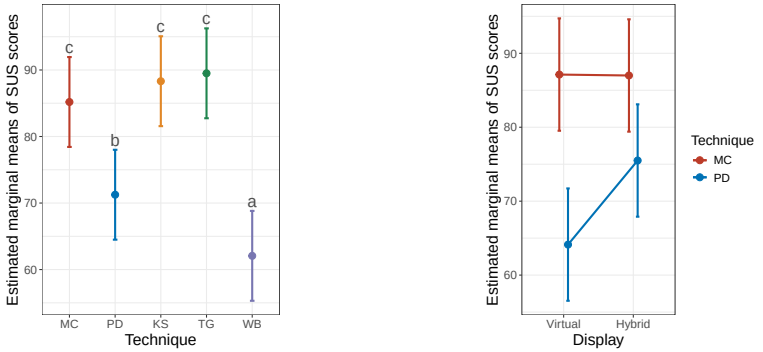
4.3 Usability

We fitted linear mixed-effects models for SUS scores, with Technique and Display as fixed effects, participants as a random intercept. In our preliminary analysis of the carryover effect, we found Condition Order significantly influenced SUS scores in structured layout task but not in freeform position task. Thus, the linear mixed-effects model for structured layout task also has Condition Order as a covariate. The effects of Technique and Display were therefore evaluated while accounting for the variability associated with condition order. The descriptive statistics of SUS scores are listed in Table 3.

4.3.1 Structured Layout Task. We found a significant main effect of Technique on SUS scores, $F(4, 160) = 32.13$, $p < .001$. The main effect of Display was not significant, $F(1, 40) = 0.99$, $p = .325$. Also, the Technique × Display interaction was not significant, $F(4, 160) = 2.41$, $p = .052$. Post-hoc comparisons using Tukey adjustment showed that MC, KS, and TG achieved higher SUS scores than

PD and WB (all $p < .001$). Additionally, PD was rated higher than WB ($p = .031$). No significant differences were observed among MC, KS, and TG. The results are visualized in Figure 4a.

4.3.2 Freeform Position Task. The linear mixed-effects model revealed a significant main effect of Technique on SUS scores, $F(1, 40) = 52.82$, $p < .001$, and a significant Technique × Display interaction effect, $F(1, 40) = 5.87$, $p = .020$. The main effect of Display was not significant, $F(1, 40) = 1.45$, $p = .236$. Follow-up pairwise comparison with Holm adjustment showed that MC was rated higher than PD in both Virtual and Hybrid conditions ($p < .001$ and $p = .002$, respectively). The analysis also showed that PD was rated higher in the Hybrid condition than in the Virtual condition ($p = .038$), while MC's ratings did not significantly differ between the two display conditions. Figure 4b illustrates the interaction effect.



(a) Main effect of Technique in SLT (b) Technique × Display interaction effect in FPT

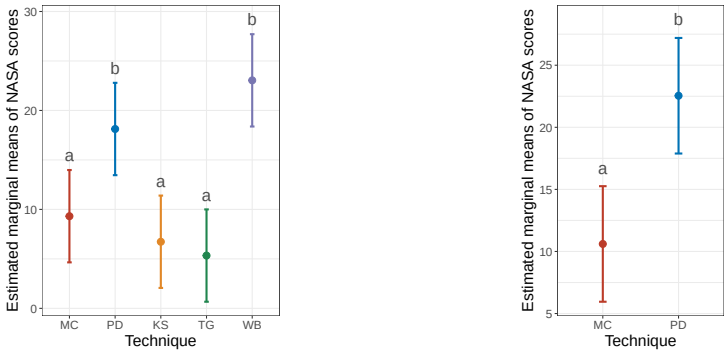
Fig. 4. Estimated marginal means of SUS scores, with error bars indicating 95% confidence intervals. (a) Structured layout task shows a main effect of Technique. A compact letter display (CLD) is used to indicate significant differences between conditions. Conditions with different letters differ significantly; shared letters indicate no significant difference (Tukey-adjusted pairwise comparisons). (b) Freeform layout task shows the Technique × Display interaction, where the effect of Technique varies across Display.

4.4 Workload

We fitted a linear mixed-effects model with Technique and Display modeled as fixed effects and participants as a random intercept. Because Condition Order did not significantly affect overall NASA scores in both tasks, it was not included as a covariate in the final models. Table 3 lists means and standard deviations of NASA scores.

4.4.1 Structured Layout Task. The model revealed a significant main effect of Technique on NASA scores, $F(4, 160) = 26.18$, $p < .001$. However, neither the main effect of Display ($F(1, 40) = 0.01$, $p = .912$), nor the Technique × Display interaction were significant ($F(4, 160) = 1.11$, $p = .354$). Post-hoc comparisons using Tukey adjustment showed that PD and WB led to significantly higher workload to complete the window placement task compared to MC, KS, and TG (all $p < .001$), as shown in Figure 5a. No significant differences were observed among MC, KS, and TG, nor between PD and WB.

4.4.2 Freeform Position Task. Similar to the previous results, the model for overall NASA scores only revealed a significant main effect of Technique, $F(1, 40) = 43.67$, $p < .001$ (see Figure 5b). Post-hoc results showed that PD received higher ratings for perceived workload than MC ($p < .001$).



(a) Main effect of Technique in SLT (b) Main effect of Technique in FPT

Fig. 5. Estimated marginal means of NASA scores across techniques. Error bars indicate 95% confidence intervals. Conditions with different letters differ significantly; shared letters indicate no significant difference. Both (a) structure layout task and (b) freeform form position task show a main effect of Technique.

4.5 Subjective Feedback

Participants' feedback was coded and categorized by Author 2 according to the reported strengths and weaknesses of the interaction techniques and the preferred technique combination. Author 1 subsequently reviewed the codes and identified themes, and both authors discussed the results to reach a shared interpretation. Figure 6 visualizes participants' feedback on their preferred technique combination. Note that five participants could not select the most preferred technique in one or both tasks, and we marked them as "no preference" in the analysis. In the structured layout task, most participants chose to use command-based techniques ($N = 28$, 70.0%), with TG being the most preferred ($N = 14$). In the freeform position task, MC was preferred; more than twice as many participants chose MC as PD. Participants' comments give more clues about these preferences.

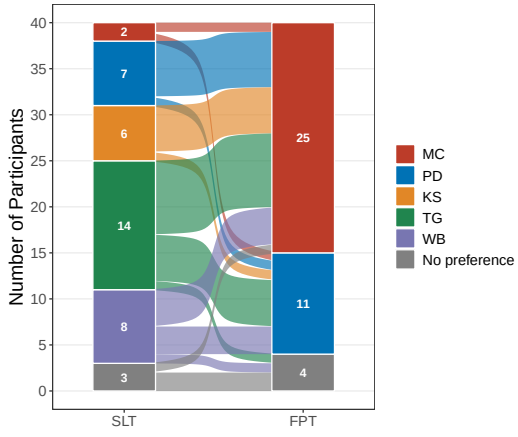


Fig. 6. Participants' preferences for window placement techniques in the structured layout tasks and freeform position tasks.

Participants were more familiar with window placement techniques adapted from a PC, including MC, KS, and TG, and consistently reported that their operations aligned with conventional practices,

leading to a low learning curve and intuitive use ($N = 18$ in total). In contrast, participants were less familiar with PD and WB, the two mid-air techniques from an HMD—they found them interesting and novel ($N = 11$), but they needed more time to master their use ($N = 13$).

Regarding accessibility and portability, P2 mentioned that it was difficult to lower his head to look at the keyboard while wearing an HMD, which makes KS error-prone. P29 worried that KS might not be used if physical keyboards no longer exist in the future workspace. On the other hand, six participants mentioned that PD and WB do not require physical input devices, allowing them to be used in more mobile scenarios. However, P7 was concerned that PD needs more space compared to other techniques and could disturb people nearby. Similarly, P35 felt embarrassed to use WB in public places. Participants did not raise any praise or concerns on this aspect in MC and TG; perhaps they are fundamental and can be used interchangeably using the integrated touchpad on a laptop.

When comparing the three shortcut techniques in the structured layout task, participants all agreed that they were fast, convenient, and easy to use. Yet six participants felt WB was error-prone, and five noted that it required bilateral coordination of both hands, which sometimes confused them.

Participants were generally satisfied with MC in both structured layout and freeform position tasks. The performance of MC was characterized by high precision ($N = 11$), high efficiency ($N = 6$), convenience ($N = 3$), and ease of use ($N = 5$). Participants have varying opinions on PD. The main advantage of using PD is that its interactions are natural and similar to those in daily life ($N = 8$). The cons of PD are twofold: first, PD requires participants to raise their hand and drag the virtual item in mid-air, which is slower and causes additional fatigue ($N = 4$); and second, some participants were concerned that hand tracking might be an issue, especially when the hand is occluded (P4, P13, and P32). Interestingly, as shown in Figure 6, MC was selected by only 2 participants in the structured layout task, but was preferred by most participants ($N = 25$, 62.5%) in the freeform position task—even including most participants who chose PD in the structured layout task. Perhaps because in the structured layout task, participants have more options from command-based techniques, while in the freeform position task, MC is faster and more stable. PD has more complaints in the freeform position task because participants felt it was difficult to move the window to an accurate position in mid-air ($N = 5$). Besides that, we did not observe any other trends in the choices made in the two tasks.

5 Discussion

This work focuses on performance and experience differences among five window placement techniques (Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, and Wrist Bending; abbreviated as MC, PD, KS, TG, and WB, respectively) across two virtual display setups (Virtual and Hybrid). All five techniques were compared in the structured layout task, while MC and PD were additionally compared in the freeform placement task.

We first discuss the results comparing the three command-based techniques (KS, TG, and WB) and the two drag-and-drop-based techniques (MC and PD). As expected, the results support the efficiency of command-based placement when the destination can be expressed as a predefined command. When using MC and PD in the structured layout task, although the snapping feature was enabled to reduce the difficulty of window placement, window navigation still took time. This may help explain why makes only about a quarter of participants prefer to use MC and PD in the structured layout task ($N = 9$). As shown in Figure 6, the preference for command-based techniques in structured layouts, together with the preference for MC in freeform placement, supports providing both approaches in XR window managers. One challenge in adapting these established approaches to 3D workspaces warrants particular attention. Suitable window positions

may depend on depth, viewing direction, available space, and the surrounding physical environment, making predefined destinations more difficult to specify and maintain as users' needs change. Future work should investigate how users can conveniently define, save, and update meaningful spatial destinations or layouts, allowing fast command-based placement to remain useful across different tasks and contexts. At the same time, predefined layouts may not accommodate the frequent adjustments needed as users shift their attention, posture, or activity. Drag-and-drop techniques should therefore support precise and comfortable manual adjustment while fitting naturally into ongoing knowledge work. Beyond reducing placement time in isolated tasks, future designs should minimize the effort of switching between window manipulation and primary work activities, allowing users to rearrange their workspace without unnecessarily interrupting their workflow.

Between the two drag-and-drop-based methods, MC was faster than PD regardless of display condition and task. This is an expected result because PD requires users to raise their hands to move the window, leading to longer strokes than MC. Moreover, participants' familiarity with PC interaction and the hand-tracking reliability of PD may also contribute to this difference. Specifically in the freeform task, the strict 0.005m placement tolerance may further amplify differences between the techniques. The requirement may have increased the need for corrective movements, particularly for PD, for which five participants reported difficulty positioning windows accurately in mid-air. The comparison therefore reflects a demanding precision task and may not fully generalize to everyday window arrangements that permit coarser placement. Future studies should vary placement tolerance or employ real-world workflow that involves coarser placement to evaluate the performance of these techniques.

On the other hand, we found that WB led to significantly more failed trials than KS and TG and was significantly slower than the other two techniques in successful repetitions in both display conditions. All these shortcut-type techniques require a special action to temporarily override the primary input's normal action. Compared to the modifier key (`Ctrl`) in KS and the additional finger in TG, the activation gesture in WB using another hand is more complex. Five participants reported confusion when coordinating both hands, which may have contributed to WB's failed trials. This bimanual activation is a special characteristic of the tested environment and implementation, and its contribution cannot be separated from that of the wrist gesture itself. However, these results do not justify dismissing WB because repeated directional placement in the experiment may have emphasized activation and coordination demands more than intermittent use would. Moreover, seven participants (the second-largest number) still preferred WB. The difference in completion time between KS and TG in the Virtual condition may be influenced by individual variability, as TG exhibited relatively higher variability in completion time. However, both techniques resulted in very short completion times (less than 1 second), indicating that the observed difference may have limited practical significance. For WB, simplifying activation could reduce bilateral coordination demands, although alternatives should be evaluated for accidental activation and mode confusion. More broadly, designing symbolic mid-air gestures for command-based window placement presents the challenge of making commands memorable, physically comfortable, and distinguishable from ordinary hand movements. Future work could explore gesture mappings that users can readily associate with placement destinations and compare activation methods while keeping these mappings constant. Longer-term studies embedded in knowledge-work tasks could further examine learnability, fatigue, and unintended activation as users alternate between window placement and their primary activities.

Regarding subjective workload and usability, we found that PD and WB both led to a significantly higher workload and lower usability than the other techniques, for which users can comfortably rest their hands on the table or laptop. Additionally, we noticed that participants often raised their hands higher than the necessary tracking height so they could see their gestures, which

might increase their workload even more. This might be because most of our participants were not very familiar with HMDs and are unclear about the HMD's hand-tracking range. Participants' substantially greater familiarity with conventional PC interaction than XR is an important qualification. Eighteen participants described the PC-derived operations as familiar and intuitive, whereas thirteen reported needing more time to master PD and WB. The results characterize use under the study's practice conditions and cannot directly be treated as estimates of performance after prolonged use. Longitudinal comparisons and recruitment across different levels of PC and XR experience could examine whether the observed differences persist as familiarity develops. PD and WB nevertheless offer possibilities that task completion time alone does not capture. Eight participants described PD as natural, and six valued the device-free use of PD and WB. These techniques may be useful when physical input devices are unavailable or for occasional window rearrangement. However, concerns about PD's space requirements and WB's use in public suggest considering the social and physical setting. An XR window manager could offer mid-air input alongside laptop techniques, allowing users to choose according to precision needs, comfort, and device availability. This complementary role warrants evaluation in realistic work sessions with both intermittent and repeated placement.

Lastly, our results showed that the Virtual and Hybrid conditions led to different performances and user experiences for some techniques. MC and PD were faster in the Hybrid condition than in the Virtual condition in both tasks, and the same for TG in the structured layout task. In addition, PD received higher usability scores in the Hybrid condition than in the Virtual condition. Unfortunately, we were unable to draw evidential conclusions on the difference between these two conditions in this study for two primary reasons. First, although we attempted to maintain consistency across the virtual windows presented in the two display conditions, achieving complete equivalence was unattainable—differences in rendering, depth perception, and the integration of virtual content with the physical environment, including the subtle differences in window sizes, movement distances, and starting surfaces (as described in Section 3.3), may have affected participants' experiences and performance, and may potentially confound the effects of display conditions. Furthermore, each participant only experienced one display setup during the experiment, which restricts opportunities to obtain direct comparative feedback about their experiences. A future study with the display setup as a within-subjects variable and seeks to match task geometry across display conditions, where practically feasible may provide further insights.

5.1 Limitations and Future Work

In this section, we discuss the limitations of this work that should be considered and point to future work.

First, we intentionally simplified the testing environment to isolate key variables and kept only the basic features of the proposed input techniques in this work. While this setup yielded rigorous findings on the use of different window placement techniques across two display setups and two tasks, future research should optimize and evaluate these techniques in a real-world context, which is often more complex and involves multitasking.

Second, as previously discussed, the Display variable was confounded with task geometry. The results derived from the Virtual and Hybrid conditions pertain to the specific configurations tested and do not establish a causal relationship regarding display setup. Future research should either control for or independently vary these geometric factors to precisely isolate the contribution of display configuration.

Third, as previously noted, we regarded the structured layout task and the freeform position task as two distinct activities, with the objective of independently evaluating the performance of techniques in these tasks. Furthermore, the structured layout task was consistently scheduled prior

to the freeform position task to streamline the study design and to reduce participants' learning curve. These configurations primarily serve experimental objectives, enabling a preliminary understanding of these techniques within low-level tasks. Future research could consider implementing both tasks in random pairs to better simulate real-world scenarios.

Fourth, to ensure applicability to current consumer devices and learnability for a wider group of users, we compared only the most fundamental unimodal inputs for window placement in the emerging cross-reality and cross-device environment. Based on the findings of these minimal interaction elements, we plan to design and develop multimodal window placement techniques and integrate them into full-featured window management systems in the future.

Finally, building on the previous points, we aim to recruit more participants and go beyond lab studies to evaluate their use in daily knowledge work in the wild. More specifically, our participants were predominantly university students, which may limit the generalizability of our findings to broader user populations. Although participants came from diverse academic disciplines, the limited number of participants within each discipline prevented a meaningful analysis of the effects of academic background on interaction performance. Furthermore, participants' prior experience with conventional PC interaction techniques may have influenced their performance and preferences. Future studies should investigate how users with different educational and professional backgrounds, technical proficiency, and familiarity with conventional interaction techniques perform with and perceive the evaluated techniques.

6 Conclusion

This work investigates how users interact with virtual displays using extended reality head-mounted displays alongside personal computers—an approach that offers greater flexibility and cost savings compared to traditional multi-monitor setups. Focusing on the fundamental task of window placement, we evaluated five interaction techniques that leverage the standard inputs of both devices: Mouse Cursor, Pinch Drag, Keyboard Shortcut, Touchpad Gesture, and Wrist Bending. Through a mixed-design user study with 40 participants, we assessed all five techniques in a structured layout task and additionally compared Mouse Cursor and Pinch Drag in a freeform placement task across fully virtual and hybrid display setups.

Across both display setups, Mouse Cursor enabled faster placement than Pinch Drag, while Keyboard Shortcut and Touchpad Gesture were faster and produced fewer failed trials than the tested Wrist Bending implementation. Mouse Cursor, Keyboard Shortcut, and Touchpad Gesture also received higher usability ratings and lower perceived workload ratings than Pinch Drag and Wrist Bending. Nevertheless, participants' preferences suggest that mid-air techniques may offer complementary value, particularly for occasional rearrangement or when physical input devices are unavailable. By systematically examining the strengths and limitations of different input modalities in this cross-device context, this work provides the first empirical foundation for understanding window placement in virtual displays. This work offers insights for designing more intuitive and efficient interactions between head-mounted displays and personal computers for future workspaces.

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A Power Analysis

Based on our study design, we chose the "`F tests`" family, the "`ANOVA: Repeated measures, within-between interaction`" test, and "`a priori`" as the type of power analysis. The table below lists the input and output parameters of this a-priori analysis. Note that, we went for a stringent standard with a 0.95 power to create balanced error control. In addition, we did not assume sphericity (ϵ=1) because it is rarely true in practice. Instead, we set an ϵ of 0.75, which could represent a moderate nonsphericity correction.

Table 4. Input and output parameters used in the power analysis.

Input Parameters	Values	Output Parameters	Values
Effect size f	0.25	Noncentrality parameter λ	17.8125000
α err prob	0.05	Critical F	2.6886915
Power (1-β err prob)	0.95	Numerator df	3.0000000
Number of groups	2	Denominator df	108
Number of measurements	5	Total sample size	38
Corr among rep measures	0.50	Actual power	0.9500783
Nonsphericity correlation ϵ	0.75		

B Supplementary Analysis of Trial Order and Direction

This section details the analysis regarding the effects of Trial Order and Direction observed when analyzing the completion time in the structured layout task (see Section 4.1)

B.1 Effects of Trial Order

Trial Order had a significant effect on completion time, $F(1, 2275.29) = 48.67$, $p < .001$. The interaction between Trial Order and Technique was also significant, $F(4, 2275.27) = 4.06$, $p = .003$, indicating that performance changes across trials differed among techniques.

Follow-up trend analyses showed that completion time significantly decreased across trials for PD ($\beta = -0.043$, [-0.076, -0.010]), KS ($\beta = -0.090$, [-0.123, -0.057]), and WB ($\beta = -0.091$, [-0.128, -0.055]), indicating performance improvements with practice. In contrast, no significant trial-order trends were observed for MC ($\beta = -0.026$, [-0.060, 0.008]) or TG ($\beta = -0.018$, [-0.051, 0.015]). These differences in learning trends indicate that techniques exhibited different rates of improvement across repeated trials.

B.2 Effects of Direction

Direction had a significant effect on completion time, $F(2, 2275.25) = 24.55$, $p < .001$. The interaction between Direction and Technique was also significant, $F(8, 2275.27) = 6.14, p < .001$, indicating that the direction of the window affected techniques differently.

To further examine the Direction × Technique interaction, pairwise comparisons with Tukey adjustment were conducted within each window direction. The overall ranking of techniques was largely consistent across directions, as shown in Table 5. The only notable variation was observed in the upward direction, where KS and TG did not differ significantly.

Table 5. Pairwise comparisons among techniques within each direction.

Direction	Significant Comparisons
Left	KS < TG < WB < MC < PD
Right	KS < TG < WB < MC < PD
Up	TG ≈ KS < WB < MC < PD

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